Shang Wu

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Education

- **Northwestern University** 09/2023 – 06/2028 (Expected)
  Ph.D. in Computer Science (Advised by Prof. Han Liu)
- **Rice University** 01/2022 – 05/2023
  M.S. in Electrical and Computer Engineering *GPA: 3.83/4.0*
- **The George Washington University** 09/2018 – 05/2021
  B.S. in Computer Science *GPA: 3.62/4.0*

Research Experience

Research Scientist Intern, ByteDance 06/2025 – 09/2025
GenAI Team, Supervisor: Dr. Zhao Song, Dr Xugang Ye

- Trained and scaled a large flow-matching-based text-to-video generation model, contributing to the development of production-level video generation systems.
- Implemented and evaluated higher-order flow matching algorithms for video generation, validating the theoretical framework proposed in the our paper "High-Order Flow Matching." **(NeurIPS 2025)**
- Investigated the impact of high-order velocity supervision (acceleration and jerk) on sampling efficiency and generation quality in large-scale video diffusion models.

Research Assistant, Magicab Lab, Northwestern University 09/2023 – Present
Video Generation Models (World Models), Supervisor: Prof. Han Liu

- Proposed a reinforcement learning framework (PhysPrompt) for text-to-video models to improve physical plausibility of generated videos in out-of-distribution domains.
- Developed a physics-consistent 4D modeling framework (Phys4D) leveraging video diffusion to enable fine-grained, temporally coherent scene understanding from monocular video.
- Designed a sparse video understanding and reasoning pipeline to enhance temporal reasoning and efficiency in large video-language models. **(CVPR 2026)**
- Developed an outlier-free SpeechLM architecture enabling fast adaptation and robust quantization for speech-language multi-modal models. **(Workshop @ ICML 2024)**

Research Assistant, EIC Lab, Rice University 05/2021 – 05/2023
Robustness and Model Compression, Supervisor: Prof. Yingyan (Celine) Lin

- Demonstrated Vision Transformers' vulnerability to adversarial attacks by designing novel patch-based attack methods (**Patch-Fool**, ICLR 2022).
- Optimized Vision Transformer attention via Taylor approximation and co-designed hardware accelerators (**ViTALiTy**, HPCA 2023).
- Developed knowledge distillation and dynamic learnable precision methods for efficient neural network compression.
- Investigated model robustness under various sparsity levels and implemented pruning methods on robust/natural models.

Research Assistant, Adaptive Devices and Microsystems Lab, GWU 01/2019 – 05/2021
Memristive Neuromorphic Computing, Supervisor: Prof. Gina C. Adam

- Implemented weight update algorithms on memristive devices and investigated training dynamics to bridge the simulation-to-device performance gap.

Publications (* denotes equal contribution)

Under Review (9 papers)

1. **Shang Wu**, Chenwei Xu, Zhuofan Xia, Weijian Li, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, and Han Liu. **"PhysPrompt: RL-based Prompt Refinement for Physically Plausible Text-to-Video Generation."** Under Review, 2026.
2. **Shang Wu***, Yen-Ju Lu*, Haozheng Luo*, Jerry Yao-Chieh Hu, Jiayi Wang, Jing Liu, Najim Dehak, Jesus Villalba, and Han Liu. **"Outlier-free SpeechLM with Fast Adaptation and Robust Quantization."** Under Review, 2026.
3. Chenwei Xu, Jianshu Zhang, **Shang Wu**, Lie Lu, Fan Du, Manling Li, and Han Liu. **"VideoCritic: Diagnosing and Localizing Reasoning Errors in Video-Language Models."** Under Review, 2026.
4. Maojiang Su, Jerry Yao-Chieh Hu, Hude Liu, **Shang Wu**, Zhao Song, and Han Liu. **"Computational Limits of Training Conditional Diffusion Transformers."** Under Review, 2026.
5. Maojiang Su, Tsung-En Lin, Jerry Yao-Chieh Hu, Guo Ye, Haoran Lu, **Shang Wu**, Zhaoran Wang, and Han Liu. **"Flow Matching Reinforcement Learning via SDE Inference."** Under Review, 2026.
6. Haoran Lu*, **Shang Wu***, Jianshu Zhang, Maojiang Su, Guo Ye, Chenwei Xu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Zhaoran Wang, and Han Liu. **"Phys4D: Fine-Grained Physics-Consistent 4D Modeling from Video Diffusion."** Under Review, 2026.
7. Haozheng Luo, Chenghao Qiu, Yimin Wang, **Shang Wu**, Jiahao Yu, Zhenyu Pan, Weian Mao, Haoyang Fang, Hao Xu, Han Liu, Binghui Wang, and Yan Chen. **"GenoArmory: A Unified Evaluation Framework for Adversarial Attacks on Genomic Foundation Models."** Under Review, 2026.
8. Guo Ye, Zexi Zhang, Xu Zhao, **Shang Wu**, Haoran Lu, Shihan Lu, and Han Liu. **"Learning to Feel the Future: DreamTacVLA for Contact-Rich Manipulation."** Under Review, 2026.
9. Maojiang Su, Mingcheng Lu, Jerry Yao-Chieh Hu, **Shang Wu**, Zhao Song, Alex Daniel Reneau, and Han Liu. **"A Theoretical Analysis of Discrete Flow Matching Generative Models."** Under Review, 2026.

Selected Published Papers (17 total)

1. Chenwei Xu, Zhen Ye, **Shang Wu**, Weijian Li, Zihan Wang, Zhuofan Xia, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, and Han Liu. **"Towards Sparse Video Understanding and Reasoning." CVPR 2026**.
2. Maojiang Su, Jerry Yao-Chieh Hu, Yi-Chen Lee, Ning Zhu, Jui-Hui Chung, **Shang Wu**, Zhao Song, Minshuo Chen, and Han Liu. **"High-Order Flow Matching: Unified Framework and Sharp Statistical Rates." NeurIPS 2025**.
3. **Shang Wu***, Yen-Ju Lu*, Haozheng Luo*, et al. **"Fast Adaptation and Robust Quantization of Multi-Modal Foundation Models from Associative Memory: A Case Study in SpeechLM."** Workshop @ **ICML 2024**.
4. Zhongzhi Yu, **Shang Wu**, Yonggan Fu, Shunyao Zhang, and Yingyan Lin. **"Hint-Aug: Drawing Hints from Foundation Vision Transformers towards Boosted Few-shot Parameter-Efficient Tuning." CVPR 2023**.
5. Yonggan Fu, Ye Yuan, Souvik Kundu, **Shang Wu**, Shunyao Zhang, and Yingyan Lin. **"Nerfool: Uncovering the vulnerability of generalizable neural radiance fields against adversarial perturbations." ICML 2023**.
6. Jyotikrishna Dass*, **Shang Wu***, Shihui Hong*, et al. **"ViTALiTy: Unifying Low-rank and Sparse Approximation for Vision Transformer Acceleration with Linear Taylor Attention." HPCA 2023**.
7. Yonggan Fu*, Shunyao Zhang*, **Shang Wu***, Cheng Wan, and Yingyan Lin. **"Patch-fool: Are vision transformers always robust against adversarial perturbations?" ICLR 2022**.
8. Yonggan Fu, Qixuan Yu, Yang Zhang, **Shang Wu**, et al. **"Drawing robust scratch tickets: Subnetworks with inborn robustness are found within randomly initialized networks." NeurIPS 2022**.
9. Yang Zhao, **Shang Wu**, et al. **"Instant-NeRF: Instant On-Device Neural Radiance Field Training via Algorithm-Accelerator Co-Designed Near-Memory Processing." DAC 2023**.

Full list with additional papers at ISCA, VLSI, DAC, tinyML, ICONS, Springer on Google Scholar.

Professional Skills

- **Programming:** Python, C++, C, Java, SQL
- **ML Frameworks:** PyTorch, TensorFlow, Jax, Huggingface, Langchain, Keras
- **Tools & Infrastructure:** Git, SLURM, AWS, Docker, Weights & Biases, Pandas, NumPy, Scikit-learn
- **High-Performance Computing:** Distributed training across multi-node GPU clusters (8×8 H100, 8×8 A100); experienced in multi-node job scheduling, FSDP/DeepSpeed parallelism, and large-scale training optimization on SLURM-managed infrastructure.

Academic Service

- **Reviewer**: ICLR (2024, 2025, 2026); NeurIPS (2023, 2024, 2025); ICML (2023, 2024, 2025, 2026); CVPR (2024, 2025, 2026); ECCV (2024, 2025)